

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Lokesh S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Lokesh S

15
20

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Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

18/2026.

COI - AT 3.

Artificial Intelligence

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Course: CSA-1708
Code

1. Introduction:

An AI agent is the one which perceives through environment for the information and uses the information to act upon it using actuators. In smart home lights, fans, etc., are the actuators and a lights glows, fan spins, AC switches on are the informations that are gathered by an AI agent.

Types of AI agent:

1. Simple reflex agent:

Simple reflex agents operate on the current task given to it by the agent. It does not remember the past actions. Easy to implement.

Working of simple reflex agent in smart home:

- * If temperature exceeds 30°C - ON AC
- * If No one is in the room - turn off lights
- * If some one enters - switch on lights & fan

2. Model based reflex agent:

It works on the futuristic actions and designs a model of the environment to store the completed task. It has complex structure

Working in smart home.

- * It remembers the completed tasks.
- * If someone enters, based on the accumulate actions it completes a necessary task.

3. Goal based Agent :

Goal based agent moves to a specific end point or goal from its initial state for its completion.

Working in smart home :

- * It ensures security.
- * It accumulates the resources for its completion.
- * It reduces power consumption.

4. Utility based agent :

It proceeds with a path which has highest utilization.

Working in smart home :

- * If a temperature is reducing, this agent computes the resources of the environment to set a correct temperature in AC to maintain the specific environment.

Conclusion :

An hybrid agent (Model based + utility based) is perfect for the smart home as model based creates a virtual model of the environment and remembers

its previous tasks, as the utility model assigns highest utility for the task.
For an smart home these two necessary features

2. Introduction.

If we have an AI based robot in a maze it does not straight away find the path by just trying different exits, rather it reaches the whole environment (maze) for an optimal path exit. So, it uses search techniques or algorithms for its traversal to find the exit.

There are 2 types of search algorithms namely:

- * Uniform cost search.
- * Iterative deepening search.

1. Uniform cost search.

It calculates the path cost for every path and proceeds with a minimum path cost path for its traversal.

How it works in maze.

If path 1 = 8, path 2 = 5, path 3 = 4.

It chooses the path 3 which has the minimum path cost of 4.

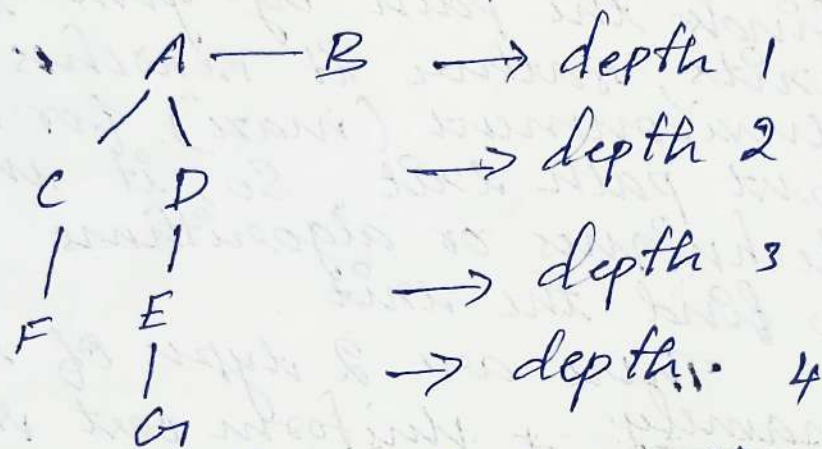
* It consecutively searches multiple paths and finds the exit.

* It ~~has~~ consumes more storage. It is the only limitation of this search.

2. Iterative deepening search:

Iterative deepening search finds the depth of every path for its traversal.

eg: If it needs to move from A to G, it calculates the distance by its depth like DFS.



* It thoroughly calculates the distance every depth, as the depth increases the time also increases.

Conclusion:

As from the analysis combining Goal based search with uniform cost search is the best option for the robot to find its path quickly. As goal based ~~search~~ Agent calculates the goal and uniform cost search calculates the path cost, combination of these 2 is the best option.